ITIL Group Reports Backend — Production Infrastructure Readiness

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Field	Value
Document reference	MOP-ITIL-REP-001
Target environment	Production
Application	ITIL Group Reports Automation — Backend and Database
Purpose	Infrastructure preparation and readiness confirmation before backend deployment
Related document	MOP-ITIL-FE-001 (frontend, Nginx, DNS, TLS)
Status	🟡 Pending — Section 2.4 host values not yet confirmed by Infrastructure

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1. Scope

This document requests only the host preparation, storage, access, network, and runtime prerequisites required before the application team deploys the release.

This backend deployment consists of:

- A FastAPI/Python application container, including the reporting pipeline and dbt.
- A PostgreSQL 16 container.
- Persistent host storage for PostgreSQL data, application data, output, and logs.

Production should mirror the proven QA container names, directory layout, and shared rootless Podman network as closely as possible. Infrastructure prepares the host; the application team owns application release deployment.

> ℹ️ **Note — integration contract:** The frontend, Nginx, DNS, TLS, host port `3001`, frontend environment variables, and frontend rollback are covered separately by **MOP-ITIL-FE-001**. This document retains only the shared integration contract: the frontend reaches the backend at `http://itil-backend-api:8000` on the rootless Podman network `itil-net`.

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2. Infrastructure actions required

2.1 Runtime and host utilities

Please install or confirm the following on the production Linux host:

- Podman 4.x or later.
- A Compose implementation compatible with `docker-compose.yml` (`podman compose` or `podman-compose`).
- `bash` 4 or later, `curl`, `tar`, `gzip`, `unzip`, and `ca-certificates`.
- `cron` / `cronie`, with the daemon enabled.
- Time synchronisation enabled and the host timezone set to `Africa/Johannesburg`.

> ℹ️ **Note:** Python, dbt, and PostgreSQL do not need to be installed directly on the host; they are supplied by the application containers.

The deployment requires the following container images. Please provide approved registry access or arrange for the images to be preloaded:

```
python:3.12-slim
postgres:16-alpine

```

2.2 Service account and access

Please provide:

- A dedicated production service account.
- Rootless Podman configuration for that account, including subordinate UID/GID mappings where required.
- User lingering enabled for that account so rootless containers survive logout and can return after a host restart.
- Ownership of the application paths listed below.

- SSH access from the approved deployment/admin workstation to TCP `22`.
- Permission to manage the application's containers and crontab.
- Limited sudo access only where required to manage the cron service or approved system services.

2.3 Persistent directories

Please create the following directory structure:

```
sudo mkdir -p /opt/itil-report-stack/scripts
sudo mkdir -p /opt/itil-report-stack/backend/releases
sudo mkdir -p /opt/itil-report-stack/backend/shared/{pgdata,backups,data,output,logs/dbt,dbt/target}
sudo chown -R svc_itil_reports:svc_itil_reports /opt/itil-report-stack
sudo chmod 750 /opt/itil-report-stack

```

> ℹ️ **Note — shared host layout:** `/opt/itil-report-stack/scripts` is shared across both applications for common operational scripts. Each application owns its own `releases/` and `current` symlink under its own subdirectory (`backend/`, `frontend/`) so the two MOPs never write to the same path — see MOP-ITIL-FE-001 for the frontend equivalent.

The following paths must persist across deployments:

```
/opt/itil-report-stack/backend/shared/.env
/opt/itil-report-stack/backend/shared/pgdata/
/opt/itil-report-stack/backend/shared/backups/
/opt/itil-report-stack/backend/shared/data/
/opt/itil-report-stack/backend/shared/output/
/opt/itil-report-stack/backend/shared/logs/
/opt/itil-report-stack/backend/shared/dbt/target/

```

2.4 Network and firewall rules

Direction	Port/protocol	Source or destination	Purpose
Inbound	TCP `22`	Approved deployment/admin workstation	Deployment and administration
Internal only	TCP `8000`	Frontend container on `itil-net`	FastAPI backend

Internal only	TCP `5432`	Backend container on `itil-net`	PostgreSQL; do not expose externally
Outbound	TCP `443` or approved Helix port	Approved BMC Helix endpoint/proxy	Report data extraction
Outbound	TCP `25` or `587`	Approved internal SMTP relay	Report and reminder email delivery
Outbound	TCP `443`	Approved container registry	Image pull/build dependencies

> ⚠️ **Warning:**
>
> - TCP `8000` and `5432` must not be exposed to user networks.
> - The frontend must use `http://itil-backend-api:8000` as its internal backend origin on `itil-net`; it must not use the public frontend hostname.

> ⚠️ **Warning — Compose gap:** The current backend repository Compose file publishes TCP `8000` on the host and does not yet declare `itil-net`. Before production deployment, the application team must update or override that Compose configuration so the backend joins `itil-net` and TCP `8000` is not externally exposed. Infrastructure should verify the resulting host listeners and firewall state before go-live.

Please complete the following values and return them to the application team:

Production item	Value
Linux hostname	
Server IP	

Rootless service account	
Approved deployment source/workstation	
Registry or offline image-transfer method	
Monitoring/alert destination	

2.5 Protected production configuration

Please create or permit the application team to create:

```
/opt/itil-report-stack/backend/shared/.env

```

> ⚠️ **Warning:** Set ownership to the production service account and permissions to `600`. Secrets must be supplied through the approved secret-management process and must not be included in email or committed to source control.

Use the following production template. Values in angle brackets must be supplied through the approved production process.

```
# Runtime, logging, and pipeline
ENVIRONMENT=production
WORKSPACE_TIMEZONE=Africa/Johannesburg
EXCEL_DROP_DIR=/app/data/excel_drop
LOG_LEVEL=INFO
LOG_FORMAT=json
LOG_MAX_SIZE=20m
LOG_MAX_FILES=5
PIPELINE_WEEKDAYS_ONLY=false
PIPELINE_SCHEDULE_HOUR=7
TEST_DATA_CLEANUP_ENABLED=false

# API and authentication
API_HOST=0.0.0.0
API_PORT=8000
API_CORS_ORIGINS=<approved-production-frontend-origin>
JWT_SECRET=
SUPER_ADMIN_EMAIL=siphamandla.mdletshe@vcontractor.co.za
SUPER_ADMIN_INITIAL_PASSWORD=TemporaryAdminPassword123!
AUTH_ALLOWED_EMAIL_DOMAINS=vodacom.co.za,vodacom.com,vcontractor.co.za
REQUIRE_DB_ON_STARTUP=true

# Bundled PostgreSQL container
POSTGRES_DB=postgres
POSTGRES_USER=itil_admin
POSTGRES_PASSWORD=<production-database-password>
DB_CONNECT_TIMEOUT=5
```

```
DB_CONNECT_RETRIES=3
DB_RETRY_DELAY_SECONDS=2
DBT_RUN_RETRIES=2
DBT_RETRY_DELAY_SECONDS=5

# BMC Helix
HELIX_BASE_URL=https://vodacom-restapi.onbmc.com
HELIX_USERNAME=<approved-production-helix-service-account>
HELIX_PASSWORD=<production-helix-password>
HELIX_VERIFY_SSL=true
HELIX_PROXY_URL=
HELIX_PAGE_SIZE=1000
HELIX_MAX_RECORDS=50000
HELIX_DATE_FIELD=Create Date
HELIX_ENRICH_CHANGE_SCHEDULES=true
HELIX_CHANGE_LOOKUP_BATCH_SIZE=50
HELIX_INCLUDE_YESTERDAY_SUCCESS_CLOSURES=true
HELIX_USE_INCREMENTAL_WATERMARK=false
HELIX_PERIOD=

# Daily Report email delivery
MAIL_PROVIDER=smtp
MAIL_FROM_ADDRESS=INTSS.ServiceDelivery@vodacom.co.za
MAIL_FROM_NAME=ITIL Daily Reports
REPORT_EMAIL_RETRY_MINUTES=15
REPORT_EMAIL_MAX_ATTEMPTS=5
REPORT_EMAIL_AUTOMATION_APPROVED=true
REPORT_EMAIL_SCHEDULER_ENABLED=true
REPORT_EMAIL_SCHEDULER_INTERVAL_SECONDS=60
REPORT_EMAIL_WEEKDAYS_ONLY=true
REPORT_EMAIL_MANUAL_SEND_ENABLED=true
REPORT_EMAIL_ALLOW_EXTERNAL=false
REPORT_EMAIL_ALLOWED_DOMAINS=vodacom.co.za,vodacom.com,vcontractor.co.
za
SMTP_HOST=<approved-production-smtp-relay>
SMTP_PORT=25
SMTP_USERNAME=
SMTP_PASSWORD=
SMTP_USE_TLS=false
SMTP_TIMEOUT_SECONDS=30

# Outstanding-task reminders
TASK_REMINDER_AUTOMATION_KEY=
TASK_REMINDER_STALE_DAYS=7
TASK_REMINDER_AUTOMATION_APPROVED=false
TASK_REMINDER_SCHEDULER_ENABLED=true
TASK_REMINDER_SCHEDULER_INTERVAL_SECONDS=60
TASK_REMINDER_WEEKDAYS_ONLY=true

```

The following values must **not** be copied from QA and must be finalised for production:

- `API_CORS_ORIGINS` — use the final production frontend URL(s), with no `localhost` or QA IP addresses.
- `POSTGRES_PASSWORD` — use a new production database password.
- `HELIX_USERNAME` and `HELIX_PASSWORD` — confirm the approved production Helix service account.

- `HELIX_PROXY_URL` — set only if the production network requires it.

2.6 Backup and rollback preparation

Infrastructure must prepare the following production controls before go-live:

- A protected backup location at `/opt/itil-report-stack/backend/shared/backups/`, owned by the production service account and not accessible to ordinary users.
- Sufficient storage for the agreed retention period.
- An automated PostgreSQL logical backup using `pg_dump`.
- A daily backup schedule under the production service account.
- A documented retention and cleanup schedule.
- Monitoring or alerting for failed backups.
- A non-destructive restore test into an isolated test database/container during the production deployment/change window.
- Release directories and a `current` symlink so the application can be rolled back to the previous approved release.
- Retention of the first verified production release as the rollback baseline for later deployments.

Recommended daily database-backup command:

```
mkdir -p /opt/itil-report-stack/backend/shared/backups
podman exec itil-postgres sh -c \
  'pg_dump -U "$POSTGRES_USER" "$POSTGRES_DB"' \
  | gzip > "/opt/itil-report-stack/backend/shared/backups/itil-reports-$(date +%Y%m%d-%H%M).sql.gz"

```

Suggested crontab entry under the production service account (adjust the time to the approved production schedule):

```
0 2 * * * /opt/itil-report-stack/scripts/backup-itil-db.sh >> /opt/itil-report-stack/backend/shared/logs/backup.log 2>&1

```

Suggested retention cleanup (adjust `<retention-days>` to the approved value once Infrastructure confirms it):

```
find /opt/itil-report-stack/backend/shared/backups/ -name '*.sql.gz' -mtime +<retention-days> -delete

```

The final backup time and retention period must follow the approved production operations standard. A backup must also run immediately before every application upgrade or database schema change.

> ⚠️ **Warning — restore procedure:** The target database must be empty or prepared under the approved restore runbook; do **not** run this example against the active production database as a test:

```
gunzip -c /opt/itil-report-stack/backend/shared/backups/<approved-
backup>.sql.gz \
  | podman exec -i itil-postgres psql \
      -U <production-database-user> \
      -d <production-database-name>

```

Application rollback procedure:

```
ln -sfn /opt/itil-report-stack/backend/releases/<previous-release> \
  /opt/itil-report-stack/backend/current

cd /opt/itil-report-stack/backend/current
podman-compose up -d --build
curl --fail http://127.0.0.1:8000/api/v1/health

```

> ℹ️ **Note:** The exact Compose command must match the implementation installed by Infrastructure (`podman compose` or `podman-compose`). Database restore must be performed only when required and under the approved production change process. An application rollback does not automatically require a database restore. For the first production deployment there is no previous production release. The initial rollback is to stop the new backend and database containers and restore data only if the approved change introduced data that must be reversed. After the first release is accepted, retain it as the baseline for subsequent backend rollbacks.

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3. Application-team deployment and validation

After Infrastructure confirms readiness, the application team will:

1. Transfer the approved release to `/opt/itil-report-stack/backend/releases/<release-tag>`.
2. Start the PostgreSQL and application containers.
3. Initialise the database schema and dbt reporting models.
4. Run the first Helix pipeline load.
5. Validate the health endpoint.

6. Configure the approved pipeline and production backup schedules.
7. Test database restore and application rollback under the approved change.

Validation commands will be run under the production service account:

```
podman --version
podman ps
podman exec itil-backend-api bash deploy/init-db.sh
podman exec itil-backend-api bash deploy/run-pipeline.sh
podman exec itil-backend-api python -c \
  "import urllib.request;
urllib.request.urlopen('http://127.0.0.1:8000/api/v1/health').read()"

```

Expected health response:

```
{
  "status": "ok"
}

```

> ℹ️ **Note:** Application deployment, database initialisation, enabling email automation, and the first production data load remain application-team/change-window activities.
